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Services & Cluster DNS

ClusterIP, NodePort, LoadBalancer, CoreDNS

Java Backend Architecture Interview Series • Tutorial Edition • Easy Diagrams & Examples

The Problem Services Solve

Pods are ephemeral — they die and get new IP addresses. If Service A stores Service B's IP and B's pod restarts, the connection breaks. A Kubernetes Service is a stable virtual IP (never changes) that load-balances across all healthy pods with a matching label.

■ **Simple Analogy:** A phone number for a department (e.g., IT helpdesk: ext. 1234). The staff (pods) behind that number change over time, but callers always dial the same number. Kubernetes Service = the stable extension number.

Service Types

Type	Reachable From	Use Case	Gets External IP?
ClusterIP	Inside cluster only	Internal microservice comms	No
NodePort	External via NodeIP:PortDev/testing, on-prem		No (uses Node IP)
LoadBalancer	External via cloud LB	Production public API	Yes (cloud-provisioned)
ExternalName	DNS alias to external UR	Reach external DB by name	No

ClusterIP Service — YAML & DNS

```
# Service definition
apiVersion: v1
kind: Service
metadata:
  name: payment-service
  namespace: production
spec:
  selector:
    app: payment # routes to pods with this label
  ports:
    - port: 80 # service port (what callers use)
      targetPort: 8080 # pod port (where app listens)
  type: ClusterIP

# Other pods reach it via DNS:
http://payment-service # same namespace
http://payment-service.production # cross namespace
http://payment-service.production.svc.cluster.local # full FQDN
```

How CoreDNS Works

CoreDNS runs as a Deployment in the kube-system namespace. Every pod gets `/etc/resolv.conf` pointing to CoreDNS. When you do a DNS lookup for 'payment-service', CoreDNS resolves it to the ClusterIP automatically.

DNS Query	Resolves To	Notes
payment-service	ClusterIP (e.g. 10.96.1.45)	Same namespace shortcut
payment-service.production	ClusterIP	Cross-namespace
payment-service.production.svc.cluster.local	ClusterIP	Fully qualified
postgres-0.postgres-svc.prod.svc.cluster.local	PodIP	StatefulSet headless DNS

LoadBalancer Service for Production

```
spec:
  type: LoadBalancer
  # On AWS: provisions an ALB/NLB automatically
  # On GCP: provisions a GLB automatically
  # Gets EXTERNAL-IP after ~30s:
  # kubectl get svc -> EXTERNAL-IP: 34.102.x.x

  # Annotation for AWS NLB instead of CLB:
  metadata:
    annotations:
      service.beta.kubernetes.io/aws-load-balancer-type: nlb
```

■ **Real-World Example:** Netflix uses Kubernetes Services with ClusterIP for all internal microservice calls. Service discovery is done entirely via DNS — no hard-coded IPs. Each service team owns their Service YAML and version it alongside their Deployment.